

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
BELAGAVI – 590 018, KARNATAKA



Shri Bhagwan Mahaveer Jain Educational & Cultural Trust ®
JAIN COLLEGE OF ENGINEERING, BELAGAVI



**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

PRE-FINAL YEAR
(2025 – 2026)

MINI-PROJECT REPORT

On

"ESP32-Based Fire Fighter Helmet Using ESP-NOW Protocol "

PROJECT GUIDE

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CERTIFICATE

This is to certify that the **Mini-Project Work** entitled “**ESP32-Based Fire Fighter Helmet Using ESP-NOW Protocol**” carried out by NAYYARA Z ASHEKHAN (2JI23EC190), SAKSHI M REVANKER (2JI23EC187), SAMARTH U NESARKAR (2JI23EC118) **and** RAHUL SAWALGI (2JI23EC095) are bonafide students of **Department of Electronics and Communication Engineering, Jain College of Engineering, Belagavi**, in partial fulfilment for the award of **Bachelor of Engineering** of the **Visvesvaraya Technological University, Belagavi** during the academic year **2025-2026**. It is certified that all corrections/suggestions indicated for project assessment have been incorporated in the report. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the Bachelor of Engineering degree.

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DECLARATION

We NAYYARA Z ASHEKHAN (2JI23EC190), SAKSHI M REVANKER (2JI23EC187), SAMARTH U NESARKAR (2JI23EC118) and RAHUL SAWALGI (2JI23EC095) students of **5th semester B.E. Electronics & Communication Engineering, Jain College of Engineering, Belagavi** hereby declare that the dissertation entitled “**ESP32-Based Fire Fighter Helmet Using ESP-NOW Protocol**” has been carried out in a batch and submitted in the partial fulfilment of the requirement for the award of Bachelor’s Degree in Electronics & Communication Engineering under Visvesvaraya Technological University, Belagavi during the academic year **2025– 26**.

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Place : Belagavi

Date :



VISION

To achieve excellence in education and research for developing globally competent , ethically sound electronic and communication engineers

MISSION

1. To provide a conducive environment for learning through structured student centric, teaching learning process
2. To nurture needs of society by infusing scientific temper in students and to grow as a centre of excellence with efficient industry-institute interaction
3. To inculcate self learning skills, entrepreneurial ability and professional ethics.

Program Educational Objectives (PEO's)

Graduates will be able to

- Contemplate real-time social problems and deliver efficient solutions
- Lead and succeed in professional careers
- Contribute through research and entrepreneurship

Program Specific Outcomes (PSO's)

Graduates in the UG program in Electronics and communication engineering will be able to

- Design, verify and develop analog and digital systems by using state of art technology to contribute to the societal needs.
- Apply the knowledge in various domains of IoT, real time systems, communication systems, VLSI and embedded systems, image and signal processing using hardware and software tools.



PROGRAM OUTCOME'S (PO'S)

Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the **engineering and management principles and apply these to one's own work, as a member and leader** in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.



Course Outcomes (CO's)

Course Outcome	Course Outcome Statement	Bloom's Level
BEC586.01	Design and develop an electronic system using modern tools by following professional ethics.	L3,L4
BEC586.02	Develop managerial, teamwork and communication skills.	L3
BEC586.03	Understand the design factors to be considered that impact society.	L2,L3

Bloom's Cognitive Levels

L1: Remembering; L2: Understanding; L3: Applying; L4: Analyzing; L5: Evaluating; L6: Creating

STRENGTH OF CO MAPPING TO PO/PSOS WITH JUSTIFICATION:

1. Slight (Low) 2. Moderate (Medium) 3. Substantial (High)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	1	3	-	-	2	-	-	-	2	2	2
CO2	-	-	-	-	-	-	-	-	3	3	2	-	-	-
CO3	-	-	-	-	-	2	2	-	-	-	-	-	-	-

CO – PO - PSO	Justification
CO1 → PO1 (3) CO1 → PO2 (3) CO1 → PO3 (3) CO1 → PO4 (1) CO1 → PO5 (3) CO1 → PO8 (2) CO1 → P12 (2) CO1 → PSO1(2) CO1 → PSO2(2)	- Students apply the electronics and communication engineering skills and design products to solve complex engineering problems. - Students gain the knowledge of modern tools and skill usage. - Students can use research-based knowledge and methods including design of experiments, analysis and interpretation to provide valid conclusions. - Students can work ethically and professionally in the industry. - Students can design their projects in analog or digital fields. - Students can choose any technology to build their projects.
CO2 → PO9 (3) CO2 → P10 (3) CO2 → P11 (2)	- Students can effectively act as an individual, and as a member or leader and work in a team. - Students can comprehend and write effective reports and make documentation with effective presentation. - Learn to management and financial skills required for the execution of project.
CO3 → PO6 (3) CO3 → PO7 (3)	- Students can able to understand the impact of professional engineering solutions in societal and environmental contexts. - Students can be able to apply reasoning of contextual knowledge to assess societal, health and safety issues.

ACKNOWLEDGEMENT

Although a single sentence hardly suffices, we would like to thank almighty God for blessing us with his grace and taking our endeavour to a successful culmination.

We express our gratitude to our guide **Dr. Nagaraj Aiholli, Associate Professor, Dept. of E&CE, JCE, Belagavi**, for his valuable guidance and continual encouragement and assistance throughout the project work. We greatly appreciate the freedom and collegial respect. We are grateful to him for discussions about the technical matters and suggestions concerned to our topic.

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We extend our sense of gratitude to **Prof.V.R.Bagali , Head, Dept. of E&CE, JCE, Belagavi**, for extending support and cooperation which helped us in completion of the project work.

We would like to express our sincere thanks to **Dr. J. Shivakumar, Principal, JCE Belagavi**, for extending support and cooperation which helped us in the completion of the project work.

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ABSTRACT

- The proposed system is an ESP32-based smart fire fighter helmet that continuously monitors temperature, humidity, toxic gas levels, and heart rate.
- ESP-NOW protocol is used to establish direct, low-latency, peer-to-peer communication between two helmets.
- Each helmet displays both local sensor data and peer helmet data on an SSD1306 OLED display.
- The system aims to improve fire fighter safety by providing early warning for heat stress, gas exposure, and physiological overload.
- The project demonstrates a low-cost, power-efficient, and infrastructure-independent solution suitable for real-time rescue operations.

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INTRODUCTION

• MOTIVATION

Fire fighters operate in hazardous environments with extreme temperatures, toxic gases, and high physical stress. Traditional helmets lack real-time safety monitoring, leading to numerous preventable injuries. A smart helmet capable of sensing hazards can significantly improve survival chances.

• PROBLEM STATEMENT

There is no affordable, wearable system that provides local and remote monitoring of vital environmental parameters and heart-rate data. Communication devices relying on Wi-Fi or cellular networks fail in collapsed structures. ESP NOW provides an ideal offline communication system.

• OBJECTIVE OF PROJECT

- Monitor temperature, humidity, gas concentration, and heart rate.
- Provide instant wireless communication using ESP NOW.
- Display both local and teammate data on OLED display.
- Improve safety and situational awareness for fire fighters.

• DESIGN APPROACH

Firmware Design & Development

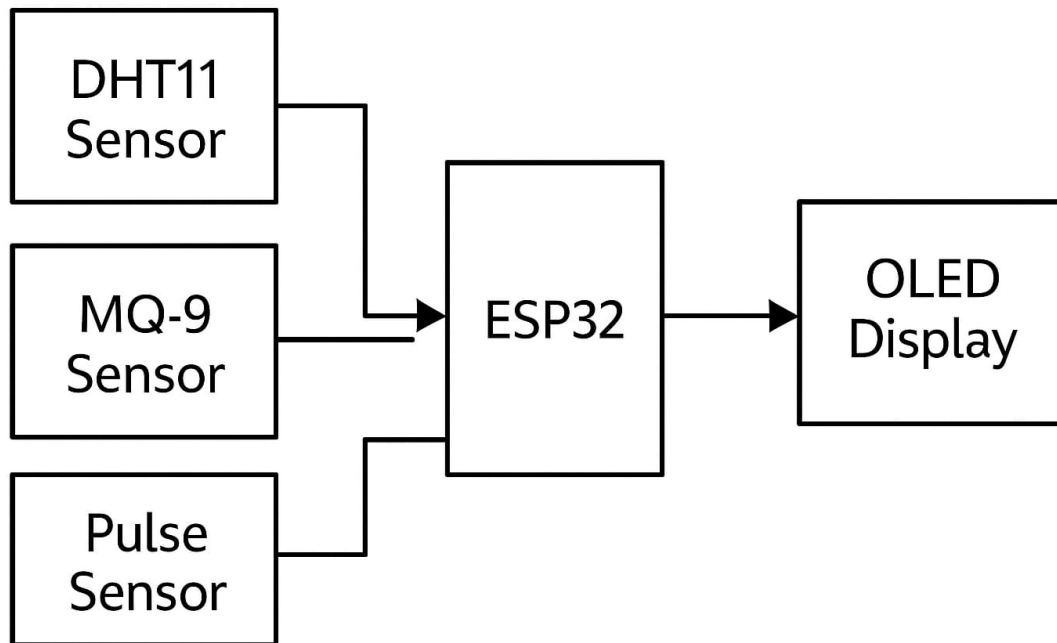
- Hardware component integration.
- firmware development.
- ESP NOW networking.
- display rendering.

Testing

- Module Testing (Hardware modules)
- Unit & Integration testing of the software
- System Testing

HARDWARE DESIGN

- **BLOCK DIAGRAM**



- **EXPLANATION OF EACH BLOCK WITH PROPER NAMES AND THEIR FEATURES AS FOLLOWS**

- **ESP32 DEVKIT V1**

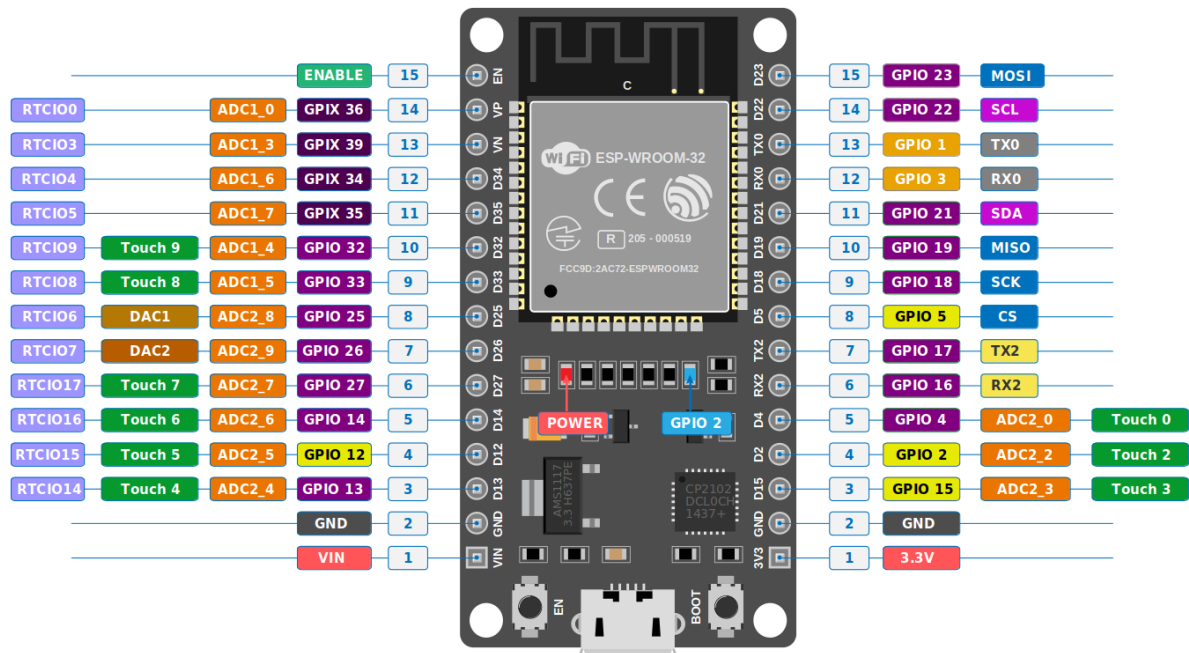


Figure 1 : ESP32 DEVKIT V1 Module with pin Description

The ESP32 DevKit V1 is a popular, cost-effective development board built around the powerful **ESP32 microcontroller**. It features integrated **Wi-Fi** and **Dual-Mode Bluetooth** capabilities, making it ideal for IoT and networking projects. The board offers a wealth of **GPIO pins**, supporting various peripherals like ADC, DAC, and SPI. Its built-in USB-to-UART converter simplifies programming and debugging, providing an easy entry point for developers.

FEATURES

- **Microcontroller:** ESP32-WROOM-32 (Dual-core processor)
- **Connectivity:** Integrated **2.4 GHz Wi-Fi** and **Dual-Mode Bluetooth** (Classic and BLE)
- **Interface:** Ample **GPIO pins** with support for ADC, DAC, I²C, SPI, and UART
- **Programming:** Easy programming via **Micro-USB** port with built-in USB-to-UART converter
- **Form Factor:** Breadboard-friendly design
- **Operating Voltage:** 3.3V

- **SSD1306 OLED DISPLAY**

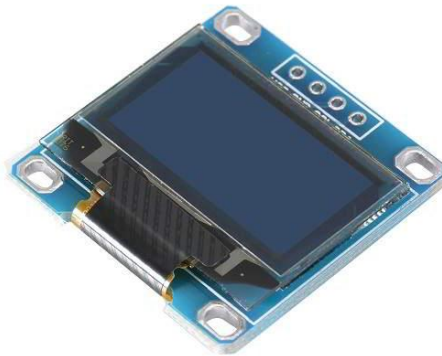


Figure 2 : SSD1306 OLED Display Module

The **SSD1306** is a widely used, low-power **OLED (Organic Light-Emitting Diode) display controller** that manages the display of pixels on small monochrome (usually white or blue) screens. These displays are popular in DIY electronics due to their **high contrast**, **fast response time**, and **low power consumption**.

FEATURES

- **Technology:** Controls a monochrome (1-bit color depth) **OLED display matrix**.
- **Resolution:** Commonly available in **128x64** or **128x32** pixel configurations.
- **High Contrast:** OLEDs emit their own light, resulting in **true black backgrounds** and excellent contrast.
- **Low Power:** Very efficient, especially when displaying fewer lit pixels.
- **Wide Viewing Angle:** OLED technology allows for a viewing angle of nearly 180 degrees.
- **Driver:** The SSD1306 driver chip handles all the necessary control and data signals, simplifying the connection to a microcontroller.

Table 1 : SSD1306 OLED Display Pin Description

SL NO	PINS	PIN DESCRIPTION
1	GND	Ground
2	VCC	Power Supply (3.3V or 5V DC)
3	SCL	Serial Clock
4	SDA	Serial Data

- **DHT11 SENSOR**

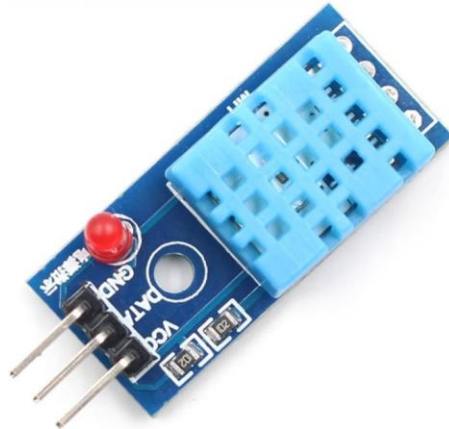


Figure 3 : DHT11 Sensor

The **DHT11** is a basic, low-cost **digital sensor** used to measure both **air temperature and relative humidity**. It integrates two sensing elements: a **resistive-type humidity sensor** and an **NTC thermistor** for temperature. An internal 8-bit microcontroller processes the analog readings and converts them into a single, calibrated **digital signal output**, which is sent over a single data line. Its simplicity and digital output make it popular for use with platforms like Arduino.

FEATURES

- Sensing Type:- Temperature & Relative Humidity (RH)
- Output Type:- Digital Signal (Single-wire serial interface)
- Temperature Range:- 0°C to 50°C (Accuracy: ±2°C)
- Humidity Range:- 20% to 95% RH (Accuracy: ±5% RH)
- Operating Voltage :- 3.3V to 5.5V DC
- Sampling Rate:- Slow (Max 1 reading per second/every ~ 2 seconds)

Table 2 : DHT11 Sensor Pin Description

SL NO	PINS	PIN DESCRIPTION
1	VCC	Power Supply (3.3V or 5V DC)
2	DATA	Digital output
3	GND	Ground

• PULSE SENSOR



Figure 4 : Pulse Sensor

The Pulse Sensor is an optical sensor using Photoplethysmography (PPG). A Green LED shines light onto the skin, and a photodetector measures the corresponding light absorbed by changes in blood volume with each heartbeat. This creates an analog signal that is amplified, allowing a microcontroller to calculate the Beats Per Minute (BPM).

FEATURES

- **Heart Rate Measurement:** Uses **Photoplethysmography (PPG)** to detect blood volume changes with each heartbeat.
- **Output:** Provides a clean **Analog Output Signal** (pulsating waveform) ready for microcontroller processing.
- **Compatibility:** Plug-and-play design, works with **Arduino**, Raspberry Pi, and other microcontrollers.
- **Operating Voltage:** Wide range, typically **3.3V to 5V DC**

Table 3 : Pulse Sensor Pin Description

SL NO	PINS	PIN DESCRIPTION
1	AOUT	Analog output
2	VCC	Power Supply (3.3V or 5V DC)
3	GND	Ground

- **MQ9 GAS SENSOR**



Figure 5 : Gas Sensor

The MQ-9 Gas Sensor is a low-cost, semiconductor-based sensor that detects both Carbon Monoxide (CO) and combustible gases like Methane (CH₄) and LPG (Propane/Butane). It works by cycling its heater voltage to measure different gases: a low temperature to sense CO and a high temperature to sense the combustible gases. This change in gas concentration is converted to a measurable analog voltage output or a simple digital alert.

FEATURES

- Main Target Gases Carbon Monoxide (CO), Methane (CH₄), LPG
- Sensing Material Tin Dioxide (SnO₂)
- Operating voltage 5.0V.

Table 4 : MQ135 Sensor Pin Description

SL NO	PINS	PIN DESCRIPTION
1	DOUT	Digital output
2	AOUT	Analog output
3	GND	Ground
4	VCC	Power supply(5.0v)

- **POWER SUPPLY**



Figure 6 : 3.7v 450mAH Li-Po Battery

Specifications:

1. **Chemistry & Voltage:** Uses **Lithium Polymer (Li-Po) chemistry** with a **3.7V nominal operating voltage** (a single cell).
2. **Capacity:** Has a charge capacity of **450 milliamp-hours (mAH)**, resulting in approximately **1.67 Watt-hours (Wh)** of energy.
3. **Charge Voltage:** The cell must be charged to a maximum voltage of **4.2V** using a dedicated Li-Po charger.
4. **Discharge Safety:** The minimum safe discharge voltage is typically **3.0V to 3.2V**; discharging below this point causes irreversible damage.
5. **Charge/Discharge Current:** Standard maximum charge/discharge rate is often **1C (450mA)**, though a safe charge rate is typically **0.5C (225mA)**.
6. **Protection:** Usually includes a **Protection Circuit Module (PCM)** to guard against overcharging, over-discharging, and short circuits.
7. **Form Factor:** **Lightweight** and **compact**, making it ideal for small, portable electronics and wearable devices.

SOFTWARE DESIGN

- **PROGRAMMING LANGUAGE**

🚦 C++

- **DEVELOPMENT TOOLS**

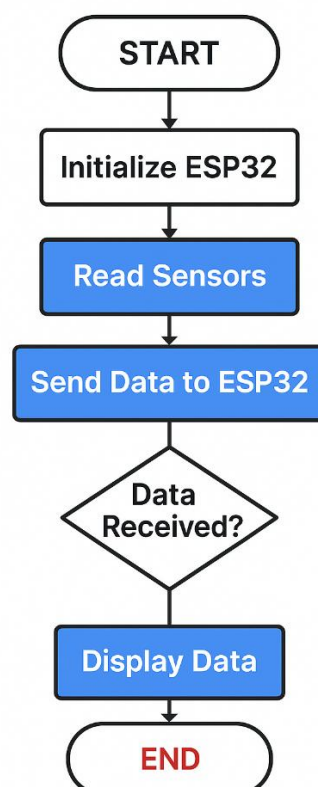
🚦 Arduino IDE

- **ARDUINO IDE**

Arduino software is needed to program Arduino boards and must be downloaded from the Arduino website and installed on a computer. This software is known as the Arduino IDE (Integrated Development Environment). Drivers must be installed in order to be able to program an Arduino from the Arduino IDE.

- **DESIGN**

1.1.1 MAIN FLOW DIAGRAM



APPLICATIONS

• APPLICATIONS

1. Firefighting Operations:- Protects firefighters from heat, falling objects, and flames.Helps detect dangerous levels of temperature and toxic gases early.Alerts the firefighter through buzzer or light signals in risky situations.
2. Real-Time Monitoring and Communication:-Sends live data (temperature, gas levels, location) to the control room.Helps in quick coordination between team members during rescue operations.Enables immediate emergency response if a firefighter is in danger.
3. Rescue and Disaster Management:-Useful in building collapse, industrial fires, or forest fires.GPS tracking helps locate trapped firefighters or victims quickly.Can support voice or message communication in smoke-filled areas.

ADVANTAGES

1. Fast, Low-Latency Communication:-ESP-NOW sends data instantly (in milliseconds).Ideal for emergency alerts when every second matters.
2. No Need for Wi-Fi or Internet:-Works without router, network, or SIM card.Perfect for operation inside buildings, tunnels, forests, or disaster zones.
3. Peer-to-Peer / Mesh Communication:-Helmets can communicate directly with each other.Can form a group network of firefighters for teamwork and coordination. Allows data relaying over longer distances.
4. Real-Time Safety Monitoring:-Integrates sensors (temperature, gas, motion, etc.)Helmet can detect danger and automatically alert others.

DISADVANTAGES

1. Most smart helmets require battery power, if battery fails during an operation the helmet stops operating.
2. Small data packets.
3. Limited number of devices can be connected.
4. Low range.
5. Current prototype does not integrate GPS or central command dashboard.

RESULTS

1. Successful Real-Time Monitoring of Environmental Parameters
2. Reliable Bi-Directional Communication Between Two Helmets Using ESP-NOW
3. OLED Display Provided Clear and Readable Output
4. Cross-Helmet Safety Awareness Achieved
5. System Operated Without Internet or Wi-Fi Router
6. System Demonstrated Low-Power Operation
7. Proof of Concept Successfully Achieved

CHAPTER 6

CONCLUSION

- The ESP32-based Fire Fighter Helmet provides a low-cost, real-time monitoring system for environmental and physiological parameters.
- ESP-NOW protocol ensures reliable, low-latency, offline communication between helmets.
- By combining gas sensing, temperature/humidity monitoring, and heart rate tracking, the system improves fire ground situational awareness.
- The helmet concept can be expanded into a larger safety ecosystem with
- Overall, the project demonstrates how embedded IoT technologies can directly contribute to fire fighter safety.

FUTURE SCOPE

- 1) Integrate GPS for location tracking of fire fighters.
- 2) Incorporate additional motion sensors for fall detection and movement analysis.
- 3) Inbuilt voice communication system like walkie-talkie.
- 4) Add central monitoring unit to aggregate data from multiple helmets.
- 5) Integration of high range antennas for better range.
- 6) Use more accurate environmental sensors (industrial-grade gas sensors, higher-resolution temperature sensors).
- 7) Implement cloud connectivity for training analysis and post-incident review.

CHAPTER 7

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